

Single Column Model (SCM) Diagnostic Runs

GSPT Simulation Pipeline

July 16, 2026

1 Overview

This document compiles every current SCM case report under results/ that has a compiled wrapper PDF.

2 Case 1: gabls1 80x2m 9h report

3 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json`

. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRE{Val{:forward}}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore. and returned Success .`

3.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 4
- Solver accepted/rejected steps: 416 / 1
- RHS evaluations: 104426
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

3.2 Forcing and Boundary Conditions

Table 1: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$8.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	$245.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\text{min,surf}}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

3.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.103721, 79.407743] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.009840, 82.319866]$

3.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

3.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

3.6 Included Plots

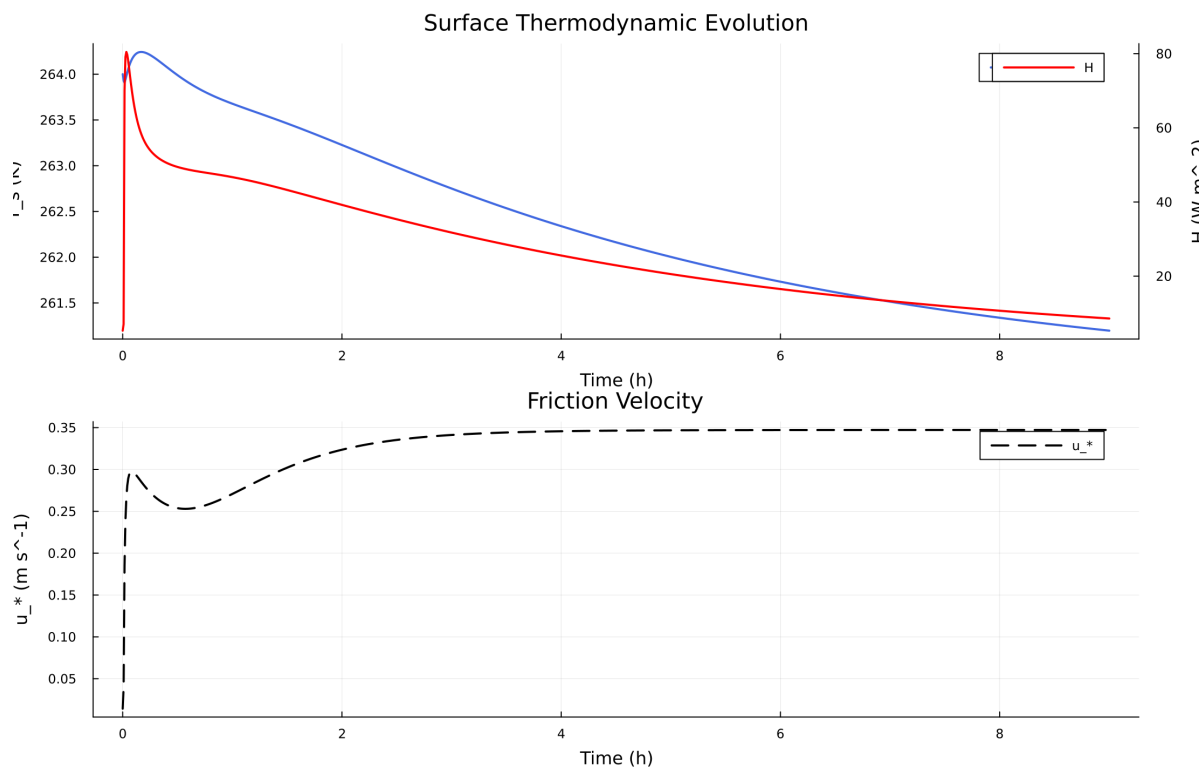


Figure 1: Time series of T_s , sensible heat flux, and friction velocity

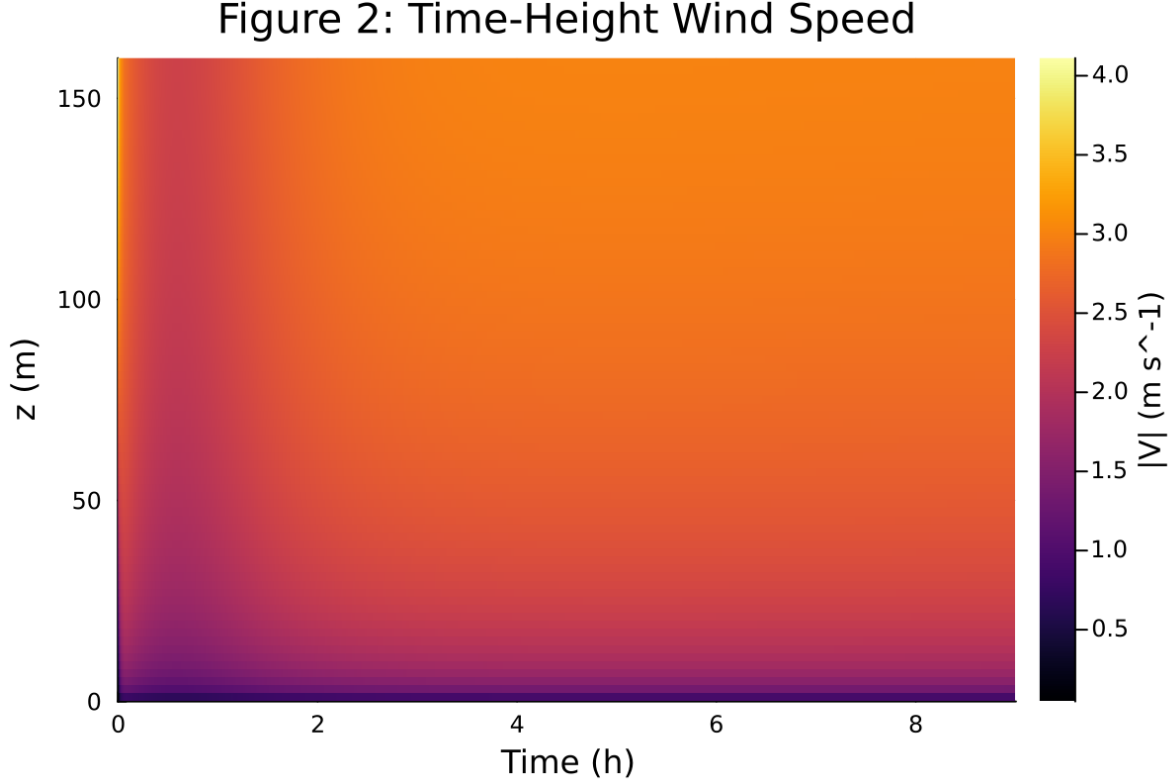


Figure 2: Time-height contour of wind speed with LLJ evolution

4 Case 2: idealized sbl 80x2m 9h report

5 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRE{Val{:forward}}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t` and returned `Success`.

5.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 4
- Solver accepted/rejected steps: 438 / 2
- RHS evaluations: 109956
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

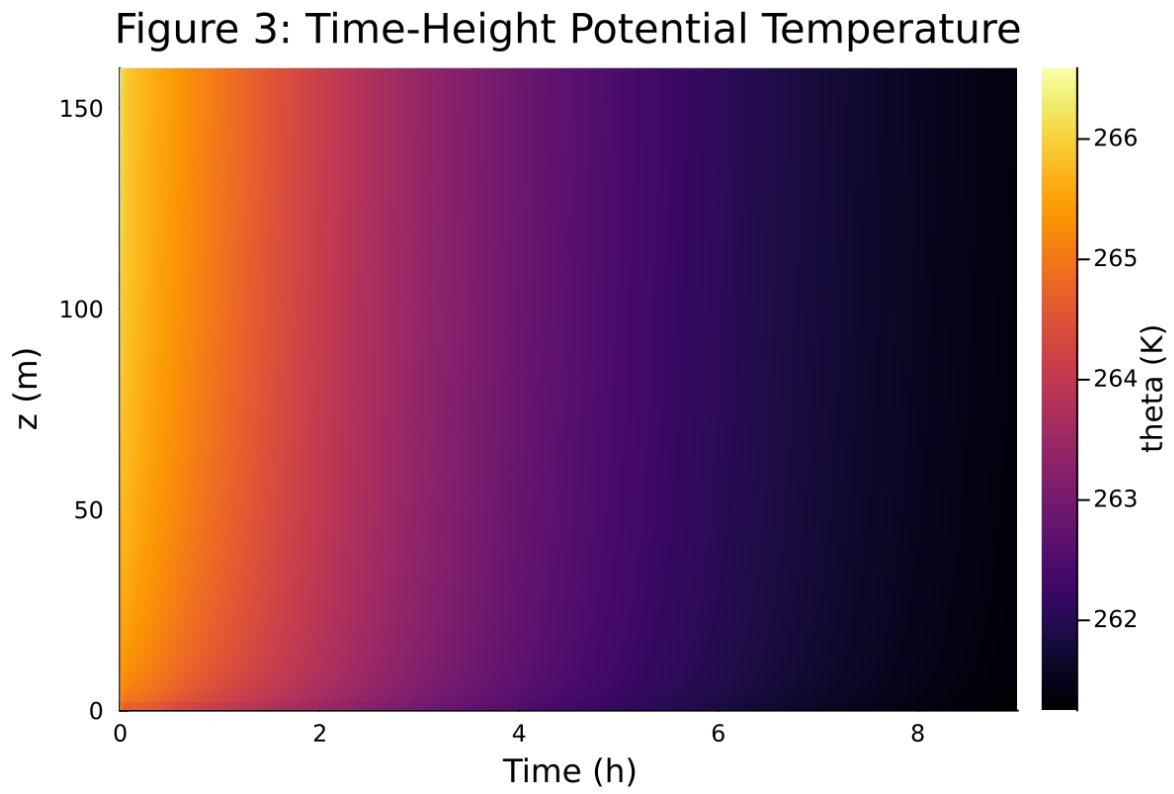


Figure 3: Time-height contour of potential temperature and inversion growth

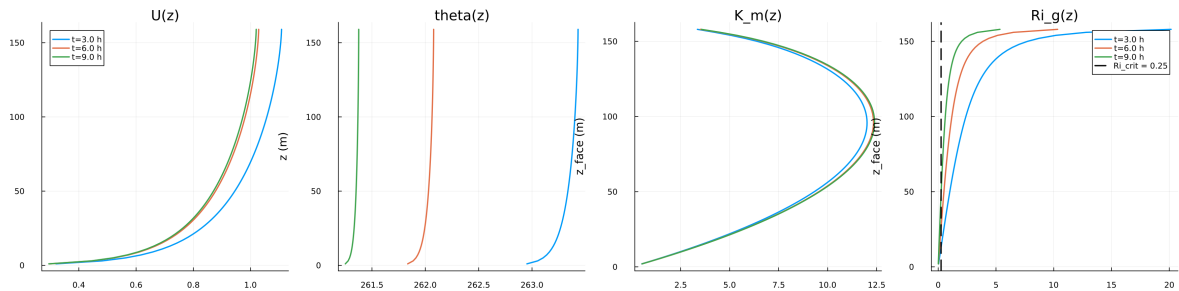


Figure 4: Vertical profiles of U , θ , and K_m at representative times

Figure 5: Surface Energy Budget

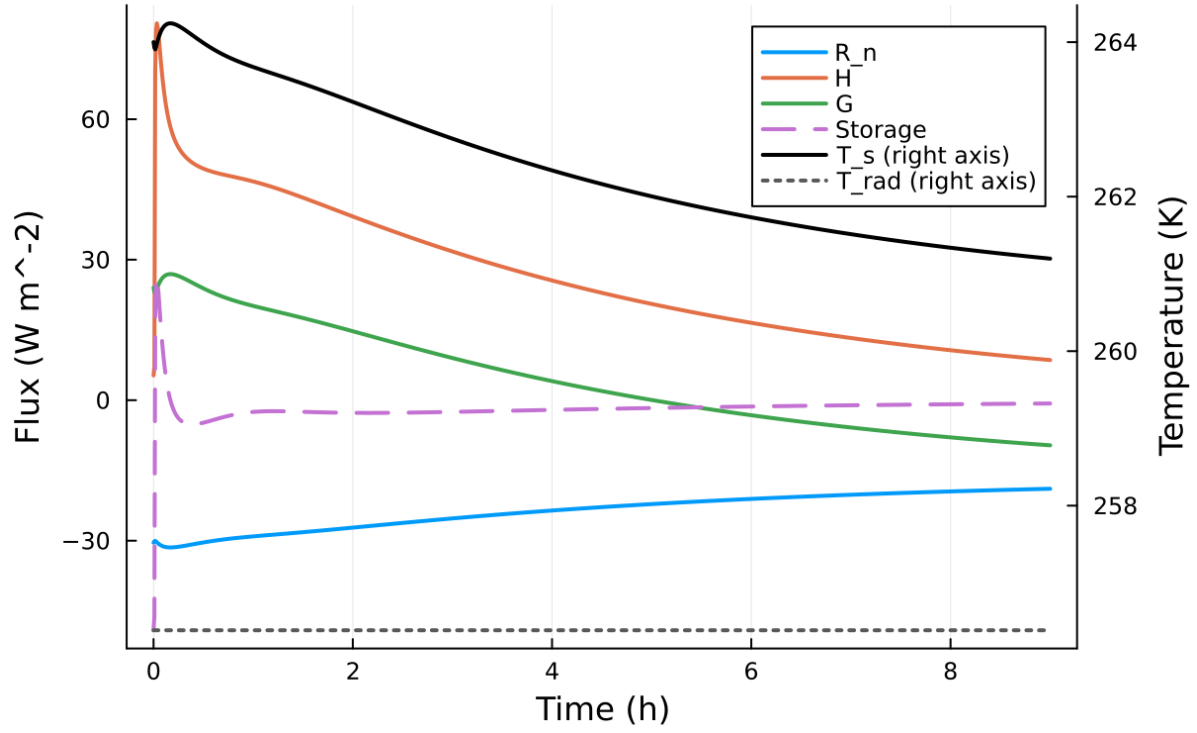


Figure 5: Surface energy budget components (R_n , H , G , storage)

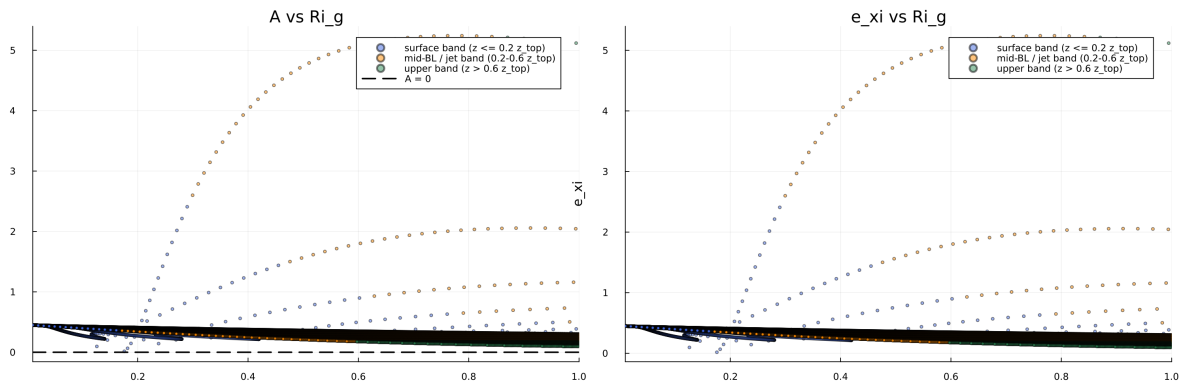


Figure 6: Phase portrait on regularized manifold (Δ vs e_{xi})

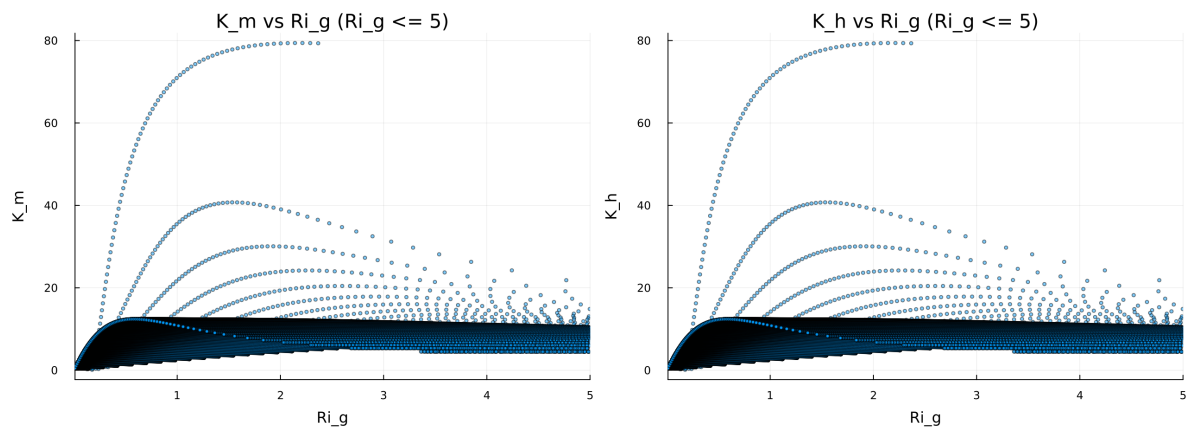


Figure 7: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

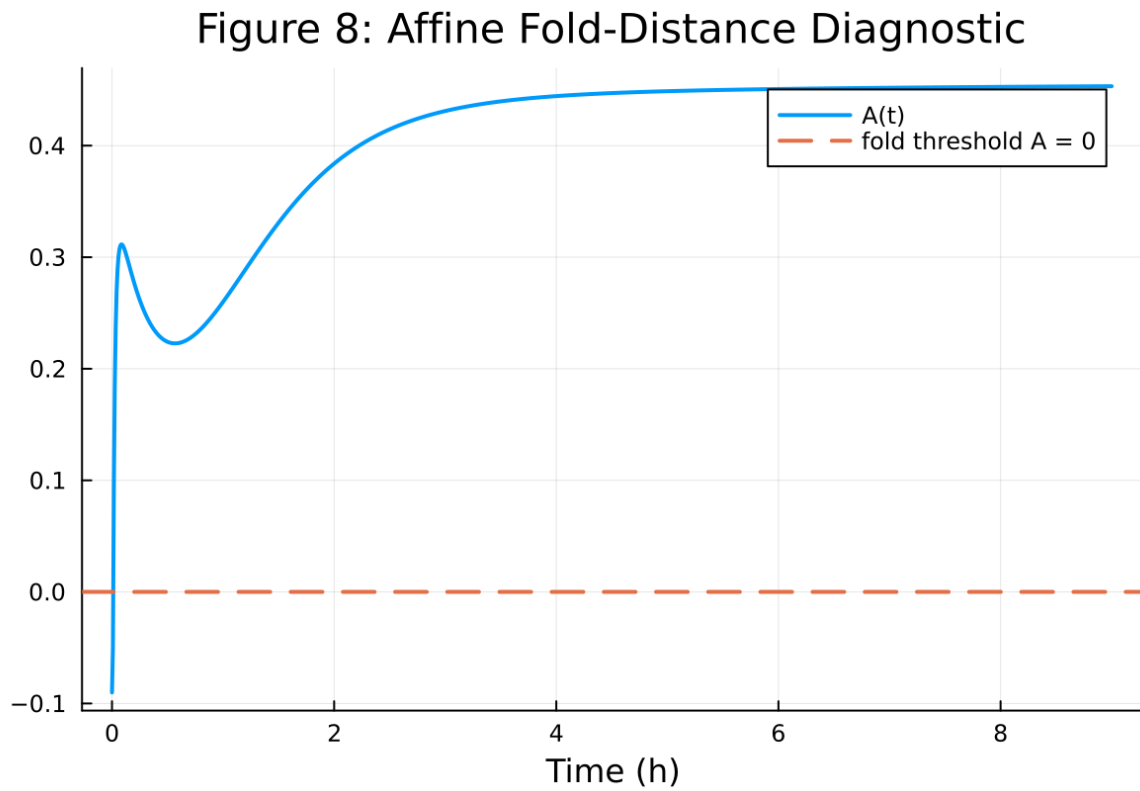


Figure 8: Fold proximity diagnostics versus time

Table 2: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\min, \text{surf}}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

5.2 Forcing and Boundary Conditions

5.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.184562, 108.426452] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.007283, 43.999865]$

5.4 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

5.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

5.6 Included Plots

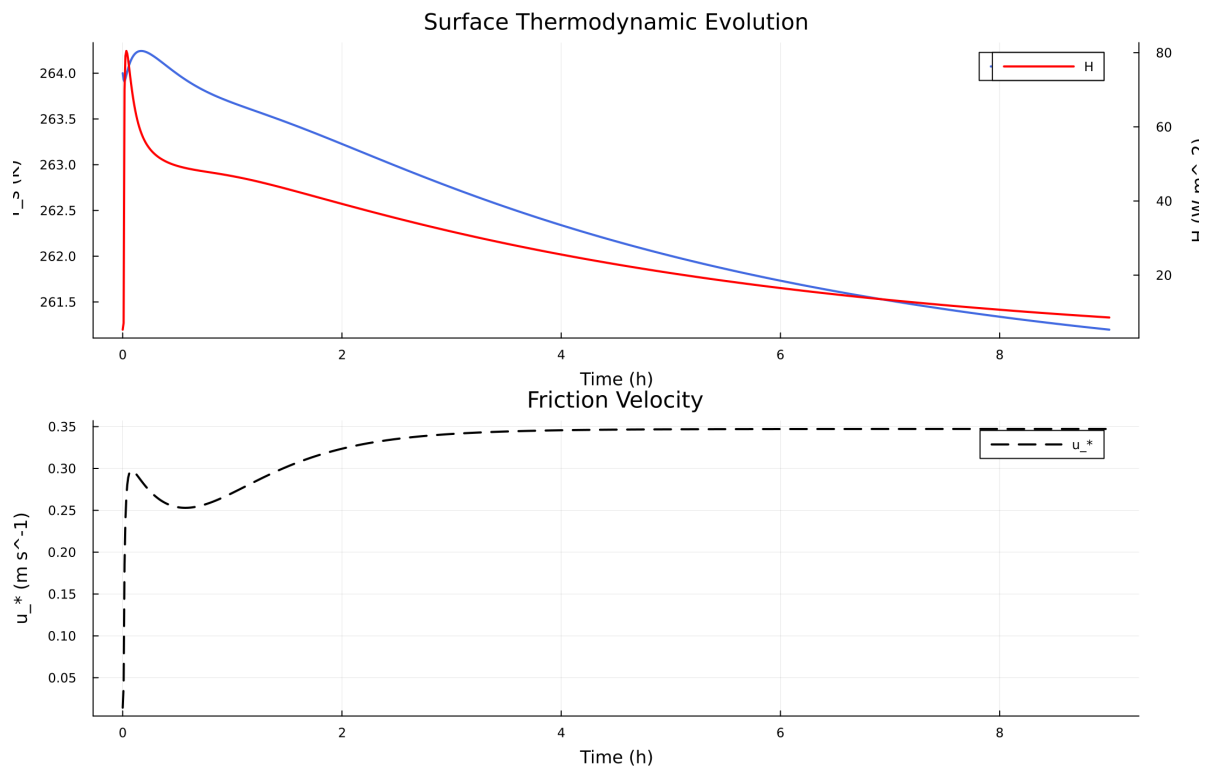


Figure 9: Time series of T_s , sensible heat flux, and friction velocity

Figure 2: Time-Height Wind Speed

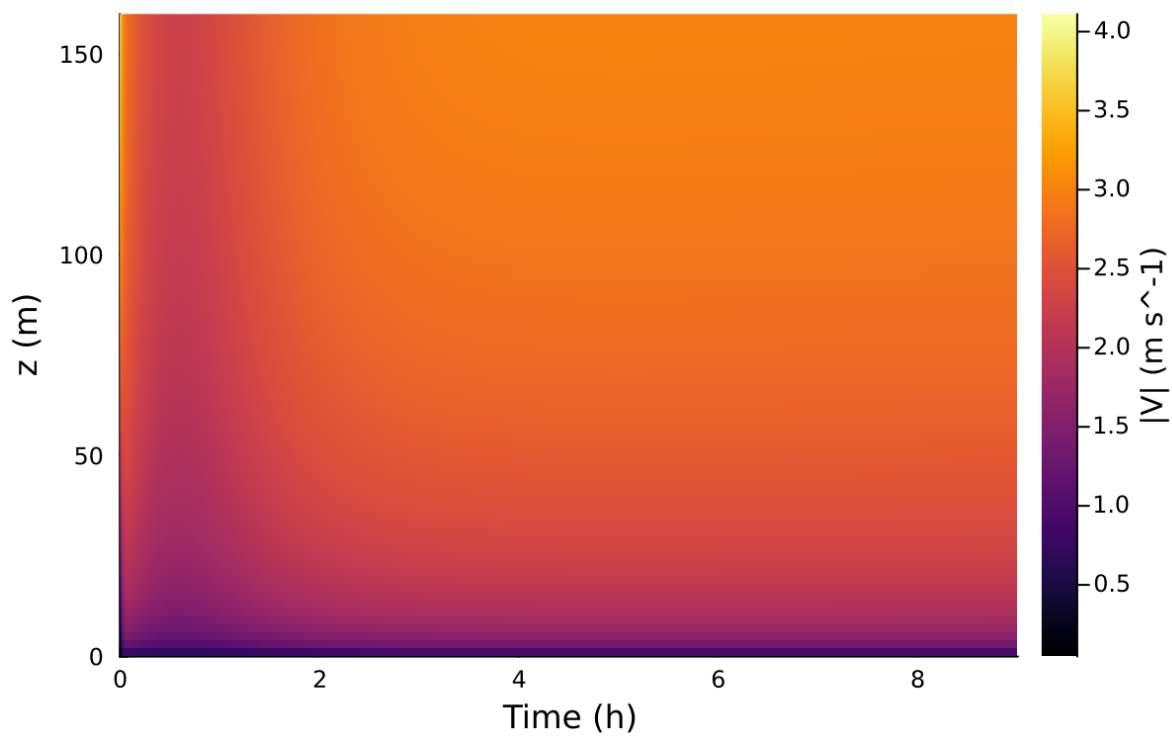


Figure 10: Time-height contour of wind speed with LLJ evolution

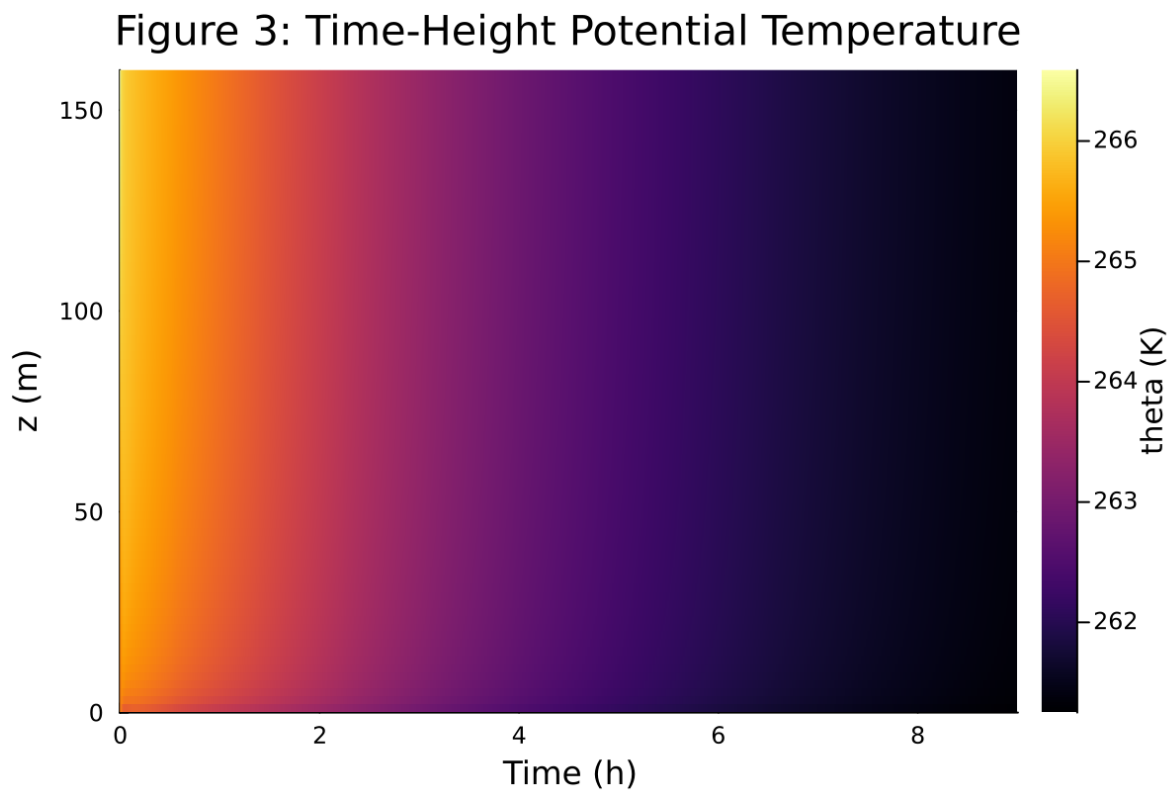


Figure 11: Time-height contour of potential temperature and inversion growth

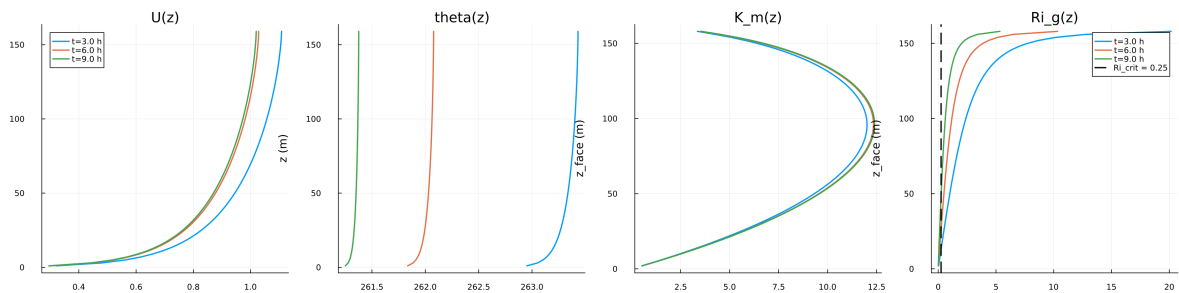


Figure 12: Vertical profiles of U , θ , and K_m at representative times

Figure 5: Surface Energy Budget

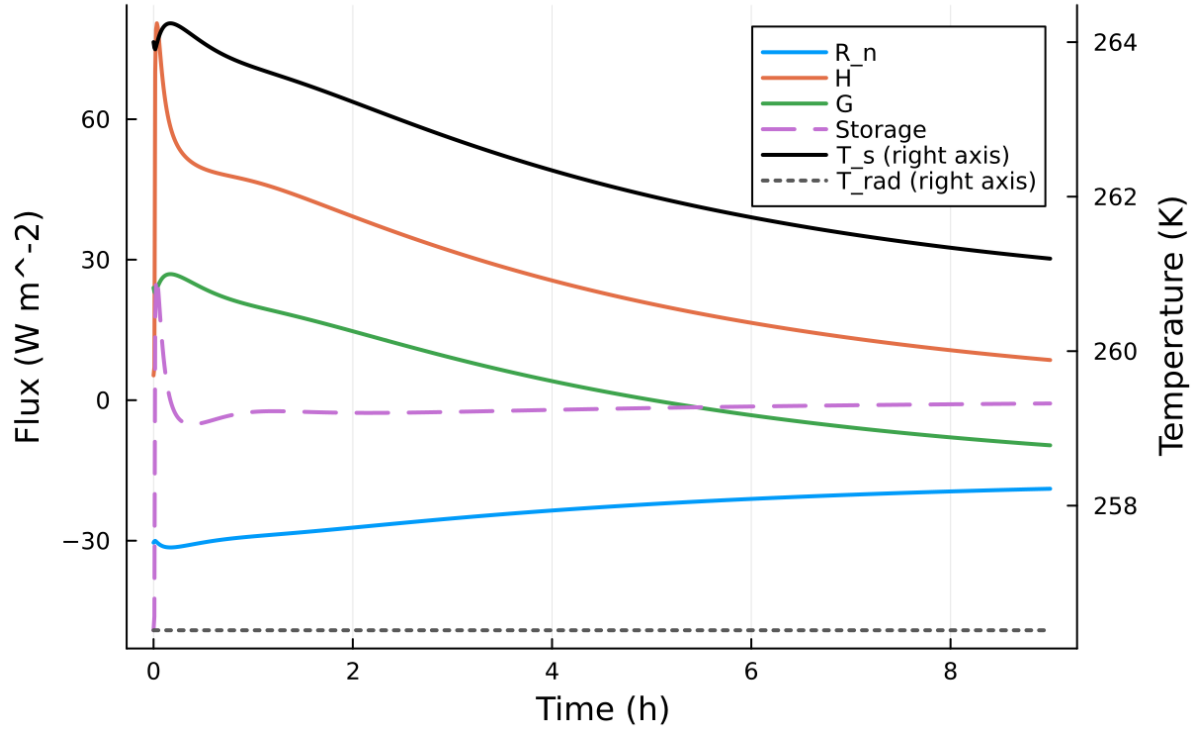


Figure 13: Surface energy budget components (R_n , H , G , storage)

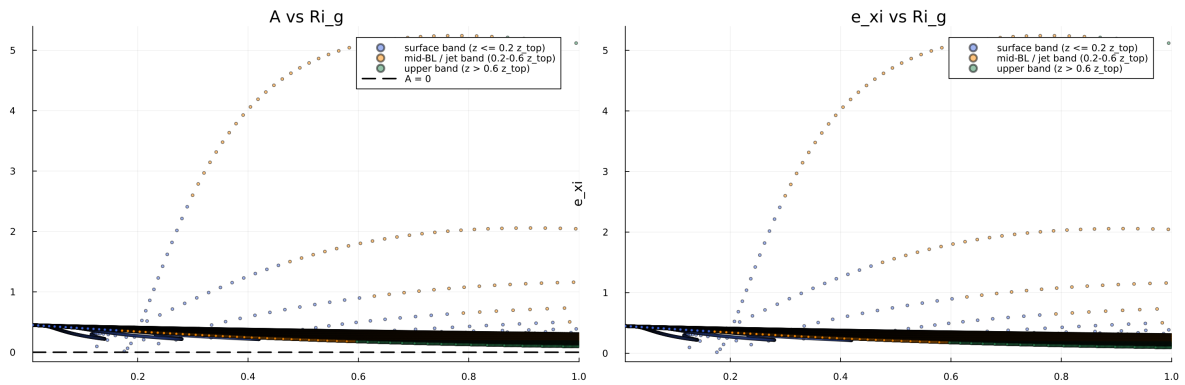


Figure 14: Phase portrait on regularized manifold (Δ vs e_{xi})

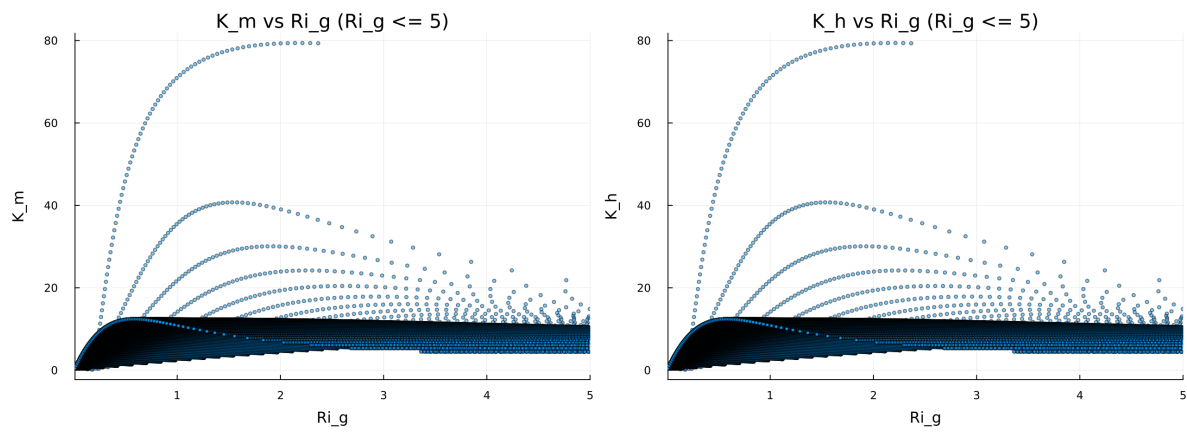


Figure 15: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

Figure 8: Affine Fold-Distance Diagnostic

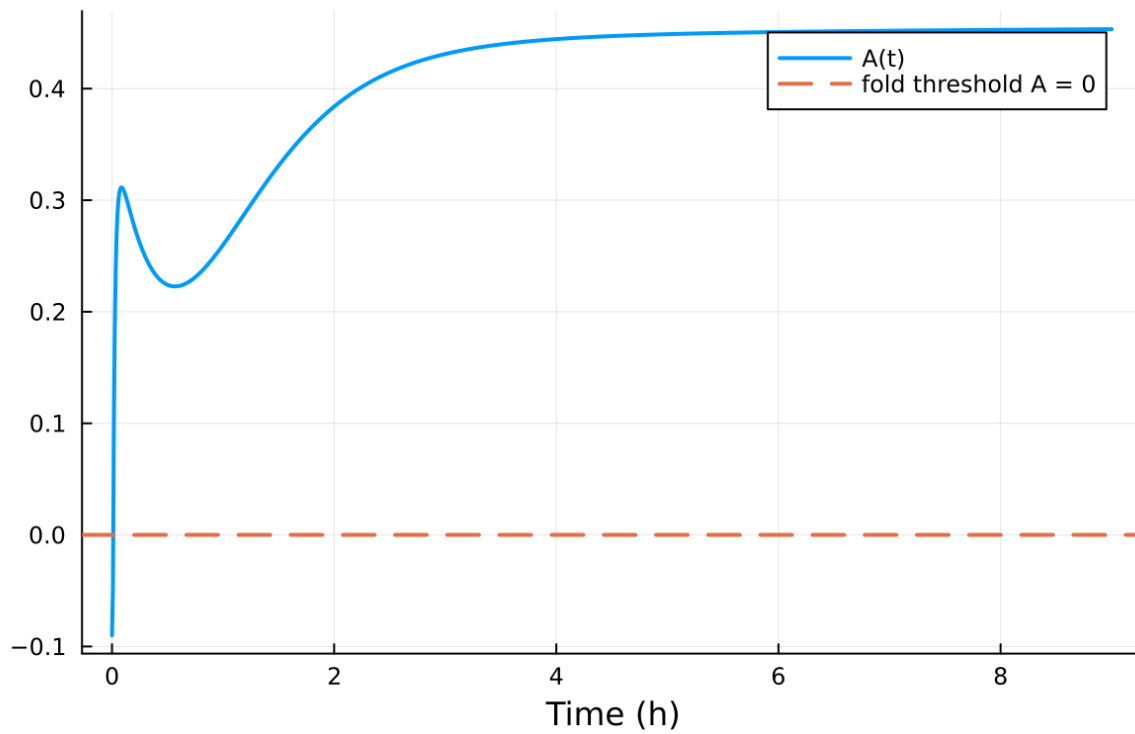


Figure 16: Fold proximity diagnostics versus time